

The gauge-free scalar potential

How Φ is recovered from the field instead of read out of the gauged system, why that makes it independent of the spanning tree, and — at least as important — what it therefore cannot do.

Implementation: `src/gauge_free_potential.cpp`,
`include/aphi_solver/gauge_free_potential.hpp`. Measurements: `GAUGE_CHOICE.md`
§15.10, `tools/gauge_spike/RESULTS_SPIKEC.md`.

0. Summary

What it is	A second, independent scalar solve that recovers Φ from E
What it fixes	R and the resistive part of a terminal voltage, at a port inside the domain, where the gauged Φ is tree-dependent by 3.1 %
What it does not fix	E , B , H , J — unchanged inputs — and the reactance , which it cannot see at all
Does tree–cotree still run?	Yes, unchanged. This is not a constraint on the gauged system; it is a separate system solved afterwards
Is it symmetric?	Yes , complex-symmetric, upper triangle only, same LDL^T as the main solve
Cost	319 ms against a 219 s factorization — 0.15 % of the run

1. The problem

Φ from the gauged $A\text{-}\Phi$ system is a **gauge representative**. Tree–cotree picks one by setting $a = 0$ on a spanning tree, which fixes ψ at every node relative to the root through

$$\psi(n) - \psi(\text{root}) = - \int_{\text{path}} A_{\text{physical}} \cdot dl$$

— a **path integral**, hence tree-dependent.

Where every terminal sits on the outer boundary this does no harm: $\mathbf{n} \times \mathbf{A} = 0$ forces $\mathbf{A}_{\text{tangential}} = 0$ there, which makes the boundary an equipotential for ψ , and `07_GaugeInvariance` measures the terminal quantities invariant to 1e-12 across three spanning trees.

Where a terminal is **inside** the domain, nothing pins ψ . `08_MixedPort_Interior` measures the consequence: the terminal inductance moves **3.1 %** and the nearby current density **77 %** with the choice of tree. `GAUGE_CHOICE.md` §15.9 then derives that **no tree can fix it** — every route to pinning ψ on an interior face either leaves it path-dependent, and so unprotected, or imposes a spurious flux constraint, and so wrong.

So Φ has to come from somewhere other than the gauged system.

2. The formulation

2.1 What is actually being computed

$\mathbf{E}$ is **gauge-invariant**. It is the one thing in the solution that does not care which tree was chosen. So: take $\mathbf{E}$ as given and ask what scalar potential it implies.

Split $\mathbf{E}$ into an irrotational part and a β -solenoidal remainder — the β -weighted **Helmholtz decomposition**:

$$\mathbf{E} = -\nabla\Phi + \mathbf{E}_{\text{sol}}, \quad \text{with} \quad \nabla \cdot (\beta \mathbf{E}_{\text{sol}}) = 0 \quad (1)$$

Φ is exactly the irrotational potential of that decomposition. Taking $\nabla \cdot (\beta \cdot)$ of (1) kills $\mathbf{E}_{\text{sol}}$ and leaves a scalar equation for Φ :

$$\nabla \cdot (\beta \nabla\Phi) = -\nabla \cdot (\beta \mathbf{E}) \quad (2)$$

equivalently $\nabla \cdot [\beta(\mathbf{E} + \nabla\Phi)] = 0$ — **find the Φ that makes $\mathbf{E} + \nabla\Phi$ divergence-free.**

$\beta = \sigma + j\omega\epsilon$ is the material weight, the same one the main assembly uses for the Φ block. The spike used $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$; the two differ by a factor $j\omega$ that appears on both sides of (2) and cancels. β was chosen so the two potentials are written in the same units and are directly comparable.

2.2 Weak form

Multiply (2) by a test function λ and integrate by parts:

$$\langle \beta \nabla \Phi, \nabla \lambda \rangle = - \langle \beta E, \nabla \lambda \rangle \quad \forall \lambda \quad (3)$$

which is what the code assembles. This is Stysch §3.1/§6.3 ([GAUGE_CHOICE.md](#) §11.8) — and the **third independent route** in that file to a unique Φ , after Chew's χ and Ostrowski & Hiptmair's electro-quasistatic gauge.

2.3 Why it is gauge-free

Not approximately, not by good conditioning — **by construction**:

The only input to (3) is E . E does not depend on the gauge. The boundary conditions do not depend on the gauge. Therefore Φ does not depend on the gauge.

No tree appears anywhere in (3). Contrast every other remedy tried: the tree contraction of §15.9 changed *which* tree, and bought 10× where nine orders were needed. This removes the tree from the question.

Relation to the gauged potential: with $E = -j\omega A - \nabla \Phi_{\text{gauged}}$, (1) gives

$$\nabla \Phi_{\text{recovered}} = \nabla \Phi_{\text{gauged}} + (j\omega A)_{\text{irrotational}}$$

so the two differ by exactly the irrotational part of $j\omega A$ — which is the gauge-dependent piece, removed.

2.4 Why there is no reactance, and why that is the same fact

(1) discards E_{sol} . The inductive contribution $-j\omega A$ is essentially solenoidal, so the decomposition assigns it to E_{sol} and it never reaches Φ .

Measured: $\text{Im}(\Phi) \approx 1\text{e-}16$ in **every** run — including case 03, where the gauged answer is entirely correct and carries $L = 22.6 \text{ nH}$.

This is not a defect to be fixed in a later version. **It is the same property that makes the method gauge-free.** The tree-dependence lives in the solenoidal part; div annihilates the solenoidal part; so the method is blind to the tree-dependence *and* to the inductance, together, for one reason.

A consequence worth stating plainly, because it bounds what any post-process could achieve: the three tree-permuted solutions differ by a field with $\text{div}(\beta \Delta E) = 0$. Any method whose only view of the solution is through a divergence is blind to that difference. Two independent attempts confirmed it — the §15.9 tree contraction (10×) and an energy-based inductance (42 % spread across three trees). The post-processing avenue is closed; correct *fields* at a mixed port require the gauge itself to change.

2.5 Provenance — the equation is published, the usage is ours

Worth separating, because the two carry different amounts of warranty.

The equation is Stysch's (6.25), verbatim up to two terms we do not have:

"Find $\phi_c \in H^1(\Omega)$ such that* $\langle \epsilon_r \text{grad } \phi_c, \text{grad } \psi \rangle + (2/r) \langle \phi_c, \psi \rangle_{\Gamma_a} = -\langle \epsilon_r E, \text{grad } \psi \rangle + j\mu_0 c k \langle g, \psi \rangle$ "

HIS (6.25)	OURS, (3)
$\langle \epsilon_r \nabla \phi_c, \nabla \psi \rangle$	$\langle \beta \nabla \Phi, \nabla \lambda \rangle$
$-\langle \epsilon_r E, \nabla \psi \rangle$	$-\langle \beta E, \nabla \lambda \rangle$
$(2/r) \langle \phi_c, \psi \rangle_{\Gamma_a}$	absent — we have no absorbing boundary
$j\mu_0 c k \langle g, \psi \rangle$	absent — his explicit source current; ours is already inside E

His (6.26) is the Lorenz-gauged variant, carrying an extra $-k^2 \langle \phi_c, \psi \rangle$ mass term; (3) is his Coulomb form, which is the MQS limit of it. He also states the well-posedness this relies on: *"The weak-form BVP (6.25) does not possess a general LF breakdown. If the boundary is not purely magnetic... the resulting system matrix always has full rank."*

Three independent sources converge on this same scalar problem — Stysch (6.25), Ostrowski & Hiptmair's electro-quasistatic gauge, and Chew's $\alpha \rightarrow 0$ limit, which [RESULTS_SPIKEC.md](#) milestone 3 measured degenerating to exactly it. That convergence is the main reason to trust the equation.

The usage is not published, and that is the part to be careful with. Stysch never solves A-Φ at all: his §3.1 explicitly chooses the "E approach" over the "A-Φ approach", so (6.25) is his formulation's **only** Φ — primary, not recovered. We instead solve a tree-gauged A-Φ

system that *already produces a Φ *, then recover a **second** one from the resulting E , and tell the reader to take R from the second and L from the first. That is his equation applied as a **repair to another formulation's output**, and it has not been found reported anywhere.

Two consequences follow, and neither is resolved:

- **No joint consistency guarantee.** His Φ and his E come out of one

formulation and satisfy one set of equations by construction. Ours come from two. Nothing guarantees that `$\Phi_{\text{recovered}}$` and the gauged E together satisfy any single system. The 0.0014 % agreement on case 03 (§5.1) is reassuring, but it is one number on one geometry.

- **It invites the obvious objection.** If a separate scalar solve is needed

anyway, why tree-gauge the first one? That *is* Stysch's argument, and it points at restructuring the formulation rather than patching it. The absence of this trick from the literature may mean it is an unremarkable shortcut, or may mean the field went the other way for a reason. **We do not know which.**

So: the formulation is published and well grounded; the application is ours and is tested on two cases.

3. Discretization

3.1 Spaces

Φ is **P1 on mesh vertices**, deliberately, although the main solve uses P2. The source `$\text{div}(\beta E)$` is piecewise constant on a tet, so P2 would double the system and buy nothing. The price is visible and small: on case 03 the recovered and gauged real parts differ by **0.0014 %**, which is that discretization gap and not a gauge effect.

E enters as `$\text{fields.e\_tet}[t]$` — one complex vector per tet, the value at the centroid, already computed by `$\text{compute\_fields}$` .

3.2 Element matrices

Per tet, with β from the tet's body and V its volume:

$$\text{stiffness} \quad K_{ij} = \beta \cdot V \cdot (\nabla \lambda_i \cdot \nabla \lambda_j)$$

$$\text{load} \quad b_i = -\beta \cdot V \cdot (E \cdot \nabla \lambda_i)$$

Both exact: $\nabla \lambda$ is constant on a tet and E is taken constant on it, so no quadrature rule is involved.

A tet with $\beta == 0$ is skipped. At DC that is every insulator, which is correct — Φ has no support there.

3.3 The system is complex-symmetric

$K_{ij} = \beta V (\nabla \lambda_i \cdot \nabla \lambda_j)$ is manifestly symmetric in i, j : β is a scalar per tet and the dot product commutes. So:

- only the **upper triangle** is stored, in `SparseSymmetricZ`. The assembly

skips `cj < ri`, and `from_pattern` **throws** on a below-diagonal entry, so an asymmetric assembly could not go unnoticed;

- it is solved with `solve_symmetric`, the project's existing complex-symmetric

`LDLT` — the same factorization the main system uses, not a second solver.

Measured residual: 1.5e-13 on case 08, 2.3e-13 on case 03.

3.4 Terminals

`build_vertex_map` mirrors the main solve's rules exactly, so that the two potentials are comparable rather than two conventions:

VERTEX	TREATMENT
on a current port	shares that port's single unknown — this is what makes a terminal a terminal
on a voltage port	prescribed at that port's value; leaves the system
on a cut	prescribed zero. Φ <i>jumps</i> across a cut, and single-valued P1 cannot represent that, so the vertex is excluded rather than quietly averaged across the jump
absent / fixed	eliminated, as in the main solve
otherwise	free

An eliminated column's known value is lifted to the right-hand side in the usual way (`rhs - = k * val`).

3.5 The reference — what fixes the additive constant

Only $\nabla\Phi$ appears in (3), so Φ is determined only up to a constant. **What pins it is the voltage port**: its vertices are prescribed, and the Dirichlet lift carries that value into the system.

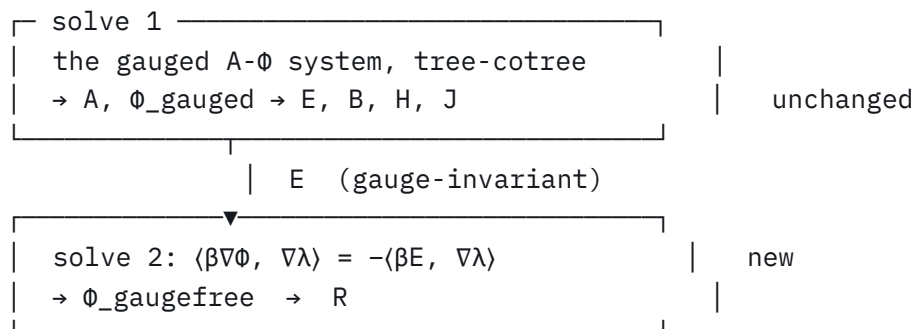
In case 08 that is `P2`, `boundary_voltage` with `voltage = 0.0` on `wire_bottom`, and the output reads back `port P2 Phi = 0 + 0j V` — exactly as prescribed, which is a free check that the elimination is right.

A reference always exists: `input_file.cpp:809` **refuses** a file whose ports are all boundary current sources, precisely because Φ would then be fixed only up to a constant. So (3) is never a pure Neumann problem in practice.

Crucially it is the **same** reference the main solve uses, which is what makes $\Phi(P1) - \Phi(P2)$ directly comparable with the gauged terminal voltage.

4. Where it sits relative to the main solve

Tree-cotree still runs, unchanged. Nothing in the gauged system was modified, no constraint was added to it, and its matrix, its gauge and its answers for `A`, `E`, `B`, `H` and `J` are exactly what they were.



Two independent linear systems. The second reads one output of the first and feeds nothing back. It can be removed without affecting anything else, and its failure cannot corrupt the main solve.

5. Measured

5.1 Validation first, on the case whose answer is already right

Case 03, both ports on the outer boundary, where the gauged answer is correct and a disagreement would mean the implementation is wrong:

	GAUGE-FREE Φ	GAUGED Φ
Re	9.79686e-05	9.796995e-05 — agree to 0.0014 %
Im	-3.0e-16 $\approx$ 0	+7.0986e-06 (L = 22.6 nH)

5.2 On the mixed interior port

Case 08, three spanning trees, against the analytic $R = L / (\sigma \cdot A_{\text{poly}}) =$
6.857804222298e-05 Ω :

	GAUGED Φ	GAUGE-FREE Φ
base	68.659272 $\mu\Omega$	68.57804222150 $\mu\Omega$
permA	68.660045	68.57804222146
permB	68.606254	68.57804222188
error vs exact	0.12 % high	2.2e-11
tree spread	7.7e-04 on R , **3.1e-02 on L **	5.5e-12
Im	+6.4e-06	-1.0e-16

About **eight orders better** on both accuracy and tree-stability.

On the eleven figures: a uniform bar has a Φ linear in z , which P1 represents **exactly**, so this geometry flatters the accuracy. Expect ordinary discretization error on a non-uniform conductor. **The tree-invariance is the geometry-independent part, and that is the result.**

6. How to use it, and how not to

QUANTITY AT AN INTERIOR MIXED PORT	WHERE TO READ IT
** R **, resistive terminal drop	** <code>potential_gaugefree.out</code> ** — exact, gauge-free
L , reactance	the gauged <code>potential.out</code> — and it is wrong by 3.1 %
B , H , local J	the field files — and they are wrong by 50–91 %

This is **not** a replacement for the gauged potential and must not be read as "mixed ports are fixed". It is a complementary extraction, reliable precisely where the gauged one is not. Both file headers carry the same rule: **read R from the gauge-free file, read L from the gauged one, and know what the latter is worth.**

7. Limits and known rough edges

- **No reactance, ever.** §2.4. Structural, not a gap to be filled later.
- **P1 against the solver's P2**, so the recovered Φ is one order lower than

the gauged one. Visible as the 0.0014 % gap on case 03.

- ** E at tet centroids** is first-order. On case 08 it still gave eleven

figures, but that is the uniform-bar effect of §5.2 and should not be expected generally. A refinement study has **not** been done.

- **A cut's Φ is excluded, not recovered.** Φ jumps across an internal cut

and single-valued P1 cannot represent the jump, so those vertices are prescribed zero. A case driven only by an internal cut therefore gets nothing useful from this file. Extending it would mean a two-sided recovery.

- ** `PhiDof::Absent` is routed through `prescribed[] = 0` **, which is *not* the

same as "has no equation". It is currently safe only because the assembly skips $\beta == 0$ tets, and absent nodes occur only where $\beta == 0$ in both regimes we run. **That is correct by accident rather than by design** and should be made explicit — either an assertion, or a branch that genuinely excludes them.

- **Not asserted in any regression test yet.** The numbers above are measured and recorded, not pinned.
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8. Bibliography

The sources this rests on, all in `APhi_Papers/`, and what each one supplies.
`GAUGE_CHOICE.md` carries the full reading of each.

Stysch (2022), *Stable Broadband Finite Element Parasitic Extraction and Sensitivity Analysis*, PhD thesis, TU Darmstadt, open access (handle [tuda/8889](#)) — [Stysch2022_PhDThesis_StableBroadbandParasiticExtraction.pdf](#)

The equation. §6.3's (6.25) is (3) up to an absorbing-boundary term and an explicit source current; (6.26) is its Lorenz-gauged variant. §6.3 also states the full-rank property (3) relies on. §3.1 is his case for solving $\mathbf{E}$ first and Φ as a post-process — and, read carefully, an argument against the way we use it (§2.5).

Stysch, Klaedtke & De Gersem (2022), *Broadband finite-element impedance computation for parasitic extraction*, *Electrical Engineering* **104**(2) 855–867, doi:10.1007/s00202-021-01348-9, preprint arXiv:2009.08232

The article form. Its §2 is the clearest published statement of why a terminal potential is gauge-dependent at all: $\mathbf{V} = \mathbf{V}_c + \mathbf{j}\omega\mathbf{I} \cdot \mathbf{L}_r$ with $\mathbf{V}_c = \Phi(\mathbf{T}_b) - \Phi(\mathbf{T}_a)$, and "as it incorporates a partial inductance, $\mathbf{V}_c$ formally depends on the gauge condition."* That is the sentence this whole document is downstream of.

Ostrowski & Hiptmair (2021), *Frequency-Stable Full Maxwell in Electro-Quasistatic Gauge*, *SIAM J. Sci. Comput.* **43**(4) B1008–B1028 (ETH SAM report 2020-43) — [ETH2020_FreqStableFullMaxwell_EQSGauge.pdf](#)

The same scalar problem reached independently, as a **gauge condition** rather than a post-process: the electro-quasistatic problem used to fix the potentials in a two-step scheme.

Chew (2014), *Vector Potential Electromagnetics with Generalized Gauge for Inhomogeneous Media: Formulation*, *PIER* **149** 69–84

The third convergence. His decoupled Φ equation (2) with $\chi = \alpha\mu\epsilon^2$ degenerates to (3) as $\alpha \rightarrow 0$, measured in [RESULTS_SPIKEC.md](#) milestone 3 — where it also shows why $\alpha \neq 0$ is unusable for a lossy conductor.

Rapetti, Alonso Rodríguez & De Los Santos (2022), *J5*(1) 52–63 —
[Rapetti2022_OnTheTreeGauge_J5.pdf](#)

Why the gauged Φ has no continuum limit to begin with: tree gauges *"are not a discretization of the Coulomb gauge and enforce no orthogonality"*, and the tree degrees of freedom are *"set arbitrarily"*. The reason §1's problem is structural rather than a matter of mesh quality.

Ansari, Farquharson & MacLachlan (2017), *GJI* **210**(1) 105–129 —
[Ansari2017_GaugedFE_CoulombLagrange_GJI210.pdf](#)

The galvanic/inductive decomposition of $\mathbf{E}$, which is the same split (1) performs — Φ carries the galvanic part, $\mathbf{A}$ the inductive. He obtains unique potentials by an explicit Lagrange-multiplier Coulomb gauge rather than by recovery, so the route differs; the decomposition does not.

COMSOL, *Computing and Using Partial Inductance with COMSOL®* (vendor documentation)

Independent public confirmation of the limit in §6: total inductance comes from stored energy, $L_{\text{tot}} = 2W_m/I^2$, while partial inductance is a free choice of subdivision and *"we can never measure any of these partial inductances, as only the total inductance of a closed loop is measurable."*

Not found reported anywhere: recovering Φ this way as a *repair* applied to a tree-gauged $\mathbf{A}$ - Φ solve, which is what this implementation does. §2.5.